

10.3

Q11

$$\sum_{n=1}^{\infty} \frac{1}{10^n} = \sum_{n=1}^{\infty} \left(\frac{1}{10}\right)^n$$

$$|r| < 1 ; \text{ as } r = \frac{1}{10}$$

So by Geometric series
it is convergent

Q12

$$\sum_{n=1}^{\infty} e^{-n} \text{ same as above}$$

Q13

$$\sum_{n=1}^{\infty} \frac{n}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{n}{n+1}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{1} = 1 \neq 0$$

So by divergent test
If the limit of n^{th}
term of sequence is
not equal to zero the
Series diverges

$$Q14 \sum_{n=1}^{\infty} \frac{5}{n+1}$$

By divergent test

$$\lim_{n \rightarrow \infty} \frac{5}{n+1} = 0$$

So test fails

we cannot apply for divergent test and Integral test

$\frac{5}{n+1}$ is positive

$\frac{5}{n+1}$ is continuous

$\frac{5}{n+1}$ is decaying

$$\begin{aligned} \int_1^{\infty} \frac{5}{x+1} dx &= 5 \ln|x+1| \Big|_1^{\infty} \\ &= 5(\ln \infty - \ln 1) \\ &= \infty \text{ diverges} \end{aligned}$$

Q15

$$\sum_{n=1}^{\infty} \frac{3}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{3}{n^{1/2}} = \sum_{n=1}^{\infty} \frac{1}{n^p}$$

by p-series test diverges
b/c $p = 1/2$

Q16

$$\sum_{n=1}^{\infty} \frac{-2}{n\sqrt{n}} = \sum a_n$$

$$\sum_{n=1}^{\infty} \frac{-2}{n^{3/2}} = -2 \sum_{n=1}^{\infty} \frac{1}{n^{3/2}} = \sum_{n=1}^{\infty} \frac{1}{n^p}$$

by p-series test $p = 3/2 > 1$
converges

Q17

$$\sum_{n=1}^{\infty} -\frac{1}{8^n} = -\sum \left(\frac{1}{8}\right)^n$$

Geometric series $r = 1/8 < 1$

So convergent.

Q18

$$\sum_{n=1}^{\infty} \frac{-8}{n}$$

Solution:

$$-8 \sum_{n=1}^{\infty} \frac{1}{n^1} = \sum_{n=1}^{\infty} \frac{1}{n^p}$$

by p-series test $p=1$

So $\sum_{n=1}^{\infty} -8/n$ diverges.

Q19

$$\sum_{n=2}^{\infty} \frac{\ln n}{n}$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \frac{\infty}{\infty}$$

D.L. Hospital Rule

$$\lim_{n \rightarrow \infty} \frac{\ln n}{1/n} = \frac{1}{1/n} = 0$$

So Divergent test fail.
OR n^{th} term test fail.

So apply integral test

$$f(x) = \frac{\ln x}{x} \quad P, C [1, \infty)$$

also decreasing

$$\frac{d}{dx} \left(\frac{\ln x}{x} \right) = \frac{\frac{1}{x} \cdot x - 1 \cdot \ln x}{x^2} < 0$$

$x \geq e$

$$\int_e^{\infty} \frac{\ln x}{x} dx = \lim_{b \rightarrow \infty} \int_e^b \frac{\ln x}{x} dx$$

$$= \lim_{b \rightarrow \infty} \left[\frac{(\ln x)^2}{2} \right]_e^b$$

$$= \lim_{b \rightarrow \infty} \left[\frac{(\ln b)^2}{2} - \frac{(\ln e)^2}{2} \right]$$

$$= \infty \quad \text{So it diverges}$$

Q19

$$\sum_{n=1}^{\infty} \frac{\ln n}{\sqrt{n}}$$

Sol apply n^{th} term test.

$$\lim_{n \rightarrow \infty} \frac{\ln n}{\sqrt{n}} = \left(\frac{\infty}{\infty} \right)$$

$$\lim_{n \rightarrow \infty} \frac{1/n}{1/(2\sqrt{n})} = \frac{2\sqrt{n}}{n} = \frac{2}{\sqrt{n}} = 0$$

So n^{th} test fails.

apply integral test

$$f(x) = \frac{\ln x}{\sqrt{x}} \quad P, C [1, \infty)$$

for decreasing $f'(x) < 0$

$$f'(x) = \frac{\frac{1}{x} \sqrt{x} - \frac{1}{2\sqrt{x}} \cdot \ln x}{(\sqrt{x})^2}$$

$$= \frac{\frac{1}{\sqrt{x}} - \frac{1}{2\sqrt{x}} \ln x}{x} \quad \text{L.C.M}$$

$$= \frac{1 - \ln x}{\sqrt{x}}$$

$$= \frac{1 - \ln x}{x \sqrt{x}} < 0$$

$$1 - \ln x < 0$$

$$1 < \ln x$$

$$e < x$$

∞

$$\int_e^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_e^b \frac{\ln x}{\sqrt{x}} dx$$

$$= \lim_{b \rightarrow \infty} \int_e^b \frac{\ln(\sqrt{x})^2}{\sqrt{x}} dx$$

$$= \lim_{b \rightarrow \infty} 2 \int_e^b \frac{\ln \sqrt{x}}{\sqrt{x}} dx$$

Let $u = \sqrt{x} \quad du = \frac{1}{2\sqrt{x}} dx$
 $\lim_{x \rightarrow \infty} u = \sqrt{x} = \infty$
 $x = \infty \quad u = \infty$

$$\lim_{b \rightarrow \infty} \int_{\sqrt{e}}^b \ln u \, du$$

$$\lim_{b \rightarrow \infty} [u \ln u - u]_{\sqrt{e}}^b$$

$$\lim_{b \rightarrow \infty} [b \ln b - b - (\sqrt{e} \ln \sqrt{e} - \sqrt{e})]$$

$$[\infty - \infty - \frac{\sqrt{e}}{2} - \sqrt{e}]$$

Q21

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n}$$

$$= \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$$

by Geometric series
 $r = \frac{2}{3} < 1$ gbr.

Q22

$$\sum_{n=1}^{\infty} \frac{5^n}{4^n + 3}$$

Solution:-

by nth term test

$$\lim_{n \rightarrow \infty} \frac{5^n}{4^n + 3} = \frac{\infty}{\infty}$$

by D.L.Hopital Rule.

$$\lim_{n \rightarrow \infty} \frac{5^n \ln 5}{4^n \ln 4}$$

$$\frac{\ln 5}{\ln 4} \lim_{n \rightarrow \infty} \left(\frac{5}{4}\right)^n = \infty$$

So by nth term test diverges.

theorem $\lim_{n \rightarrow \infty} x^n = 0$
 if $|x| < 1$
 other with $|x| > 1$ do

Q23

$$\sum_{n=1}^{\infty} \frac{-2}{n+1} = -2 \sum_{n=1}^{\infty} \frac{1}{n+1}$$

Solution:-

$$f(x) = \frac{1}{x+1}$$

Do your self.

it holds PCD

$$\begin{aligned} \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x+1} dx &= \lim_{b \rightarrow \infty} [\ln(x+1)]_1^b \\ &= \lim_{b \rightarrow \infty} \{ \ln(b+1) - \ln 1 \} \\ &= \infty \text{ diverges} \end{aligned}$$

Q24 $\sum_{n=1}^{\infty} \frac{1}{2n-1}$

similar to 23

Solution:-

$$f(x) = \frac{1}{2x-1}$$

Q25

$$\sum_{n=1}^{\infty} \frac{2^n}{n+1}$$

Solution

$$\lim_{n \rightarrow \infty} \frac{2^n}{n+1} = \frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{2^n \ln 2}{1} = 2^\infty \ln 2 = \infty \neq 0$$

So by n th term test it is dgt.

Q26

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n$$

Solution:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

So by n th term test
it divergent.

Q28

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$$

decompose

$$\frac{1}{\sqrt{n}(\sqrt{n}+1)} = \frac{1}{n(n+1)}$$

$$\frac{1}{n(n+1)} = \frac{+1}{n} + \frac{-1}{n+1}$$

Telescopic (we have practice
it in class)

Q27

$$\sum_{n=2}^{\infty} \frac{\sqrt{n}}{\ln n}$$

apply n th terms
test. (Do your self)

$$Q \sum \frac{1}{\sqrt{n}(\sqrt{n}+1)} = \sum a_n$$

$$f(x) = \frac{1}{\sqrt{x}(\sqrt{x}+1)}$$

P.C.D.

$$\lim_{b \rightarrow \infty} \int_1^b \frac{1}{\sqrt{x}(\sqrt{x}+1)} dx$$

$$= 2 \lim_{b \rightarrow \infty} \int_1^b \frac{\frac{1}{2\sqrt{x}}}{(\sqrt{x}+1)} dx$$

$$= 2 \lim_{b \rightarrow \infty} \ln(\sqrt{x}+1) \Big|_1^b$$

$$= 2 \lim_{b \rightarrow \infty} [\ln(\sqrt{b}+1) - \ln 2]$$

Example:-

Q 29

$$\sum_{n=1}^{\infty} \frac{1}{(\ln 2)^n} = \sum_{n=1}^{\infty} \left(\frac{1}{\ln 2}\right)^n$$

$$\frac{1}{\ln 2} > 1$$

So it is divergent

Q 30 =
$$\sum_{n=1}^{\infty} \frac{1}{(\ln 3)^n}$$

As Q 29

Q 31 =
$$\sum_{n=3}^{\infty} \frac{1/n}{(\ln n) \sqrt{\ln^2 n - 1}}$$

$$f(x) = \frac{1/x}{\ln x \sqrt{\ln^2 x - 1}}$$

it is P, C, D

find $f'(x)$ (yourself)

Q 31 Remaining

$$\sum_{n=3}^{\infty} \frac{1/n}{(\ln n) \sqrt{\ln^2 n - 1}}$$

$$f(x) = \frac{1/x}{\ln x \sqrt{\ln^2 x - 1}}$$

$$\int_3^{\infty} f(x) = \lim_{b \rightarrow \infty} \int_3^b \frac{1/x}{\ln x \sqrt{\ln^2 x - 1}} dx$$

Let $u = \ln x$ $\left| \begin{array}{l} x = \infty \\ u = \infty \\ x = 3 \\ u = \ln 3 \end{array} \right.$
 $du = \frac{1}{x} dx$

$$= \lim_{b \rightarrow \infty} \int_{\ln 3}^b \frac{du}{u^2 \sqrt{u^2 - 1}}$$

$$= \lim_{n \rightarrow \infty} \left[\sec^{-1} u \right]_{\ln 3}^b$$

$$= \lim_{n \rightarrow \infty} [\sec^{-1} b - \sec^{-1} \ln 3]$$

$$= \lim_{n \rightarrow \infty} [\sec^{-1} \infty - \sec^{-1} \ln 3]$$

Q32

$$\sum_{n=1}^{\infty} \frac{1}{n(1+\ln^2 n)}$$

Sol By integral test

$$f(n) = \frac{1}{n(1+\ln^2 n)}$$

$$\int_1^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x(1+\ln^2 x)} dx$$

Let $u = \ln x$ $\begin{matrix} x=\infty \\ u=\infty \\ x=1 \\ u=\ln 1=0 \end{matrix}$
 $du = \frac{1}{x} dx$

$$= \lim_{b \rightarrow \infty} \int_0^{\infty} \frac{du}{1+u^2}$$

$$= \lim_{b \rightarrow \infty} \tan^{-1} u \Big|_0^{\infty}$$

$$= \lim_{b \rightarrow \infty} (\tan^{-1} b - \tan^{-1} 0)$$

$$= \tan^{-1} \infty - 0 = \pi/2$$

Convergent

Q33

$$\sum_{n=1}^{\infty} n \sin \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = 1$$

diverges

$$\sum_{n=1}^{\infty} n \tan \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \frac{\tan \frac{1}{n}}{\frac{1}{n}} =$$

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n} \cos \frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} \times \lim_{n \rightarrow \infty} \frac{1}{\cos \frac{1}{n}}$$

$$1 \times 1 = 1 \text{ diverges}$$

Q35

$$\sum_{n=1}^{\infty} \frac{e^n}{1+e^{2n}}$$

$$\lim_{n \rightarrow \infty} \frac{e^n}{1+e^{2n}} =$$

$$f(u) = \frac{e^u}{1+e^{2u}}$$

$$\lim_{b \rightarrow \infty} \int_1^b \frac{e^u}{1+(e^u)^2} du = \lim_{b \rightarrow \infty} [\tan^{-1} e^u]_1^b$$

$$= \lim_{b \rightarrow \infty} (\tan^{-1} b - \tan^{-1} e)$$

$$= \lim_{b \rightarrow \infty} \tan^{-1} \infty - \tan^{-1} e$$

 $\pi/2 - \tan^{-1} e$ converges

Q36

$$\sum_{n=1}^{\infty} \frac{2}{1+e^n}$$

$$f(x) = \frac{2}{1+e^x}$$

$$\lim_{b \rightarrow \infty} \int_1^b f(x) dx = \lim_{b \rightarrow \infty} \int_1^b \frac{2}{1+e^x} dx$$

$$u = e^x$$

$$du = e^x dx$$

$$\frac{du}{u} = dx$$

$$\lim_{b \rightarrow \infty} \int_e^b \frac{2}{u(1+u)} du$$

$$2 \lim_{b \rightarrow \infty} \int_e^b \left(\frac{1}{u} + \frac{-1}{1+u} \right) du$$

$$2 \lim_{b \rightarrow \infty} \left[\ln u - \ln(1+u) \right]_e^b$$

$$2 \lim_{b \rightarrow \infty} \left[\ln \left(\frac{u}{1+u} \right) \right]_e^b$$

$$= 2 \lim_{b \rightarrow \infty} \left[\ln \frac{b}{1+b} - \ln \frac{e}{1+e} \right]$$

$$= 2 \left[\ln \infty - \ln \frac{e}{1+e} \right]$$

$$= \infty \text{ diverges}$$

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$$\sum_{n=1}^{\infty} \frac{8 \tan^{-1} n}{1+n^2}$$

$$f(x) = \frac{8 \tan^{-1} x}{1+x^2} \quad P, C, D$$

$$8 \int \frac{\tan^{-1} x}{1+x^2} dx$$

$$8 \left(\frac{\tan^{-1} x}{2} \right) = \frac{4 (\tan^{-1} x)^2}{1}$$

$$= 4 \left[(\tan^{-1} \infty)^2 - (\tan^{-1} 1)^2 \right]$$

$$= 4 \left[\left(\frac{\pi}{2} \right)^2 - \left(\frac{\pi}{4} \right)^2 \right]$$

Converges

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$$\sum_{n=1}^{\infty} \frac{n}{1+n^2}$$

Err Sol

$$f(x) = \frac{x}{1+x^2} \quad P, C, D$$

$$\lim_{b \rightarrow \infty} \int_1^b \frac{x}{1+x^2} dx$$

$$\frac{1}{2} \lim_{b \rightarrow \infty} \int_1^b \frac{2x}{1+x^2} dx$$

$$= \frac{1}{2} \lim_{b \rightarrow \infty} \ln |1+x^2| \Big|_1^b$$

$$= \frac{1}{2} \lim_{b \rightarrow \infty} \left[\ln |1+b^2| - \ln |1+1^2| \right]$$

$$= \frac{1}{2} \left[\ln(1+\infty) - \ln 2 \right] = \infty$$

